Remarks on Hörmander L^2-existence thm

1) $L' = K_M \otimes L = \Omega^{n,0} \otimes L$ with metric induced by ω, h_L

has curvature $\Omega_{L'} = \Omega_L - \mathrm{Ric}$

$\Rightarrow$ if $\boxed{\Omega_L \geq c\omega}$, then

$$\Omega_{L'} + \mathrm{Ric} \geq c\omega$$

So given $\alpha \in \Omega^{n,1}(L)$ s.t. $\bar\partial\alpha = 0$

$\exists\ f \in \Omega^{n,0}(L)$ s.t. $\bar\partial f = \alpha$

& $\|f\|_{L^2} \leq \|c^{-1/2}\alpha\|_{L^2}$

If $\exists\ s \in H^0(M, K_M)$ with $C_1^{-1} < |s|^2 < C_1$

then given $\alpha \in \Omega^{0,1}(L)$ s.t. $\bar\partial\alpha = 0$,

we have $\alpha' = \alpha \otimes s \in \Omega^{n,1}(L)$, $\bar\partial\alpha' = 0$

so $\exists\ f' \in \Omega^{n,0}(L)$ s.t. $\bar\partial f' = \alpha'$

& $\|f'\|_{L^2} \leq \|c^{-1/2}\alpha'\|_{L^2}$

$f = f'/s \in \Omega^0(L)$ satisfies $\bar\partial f = \alpha$

& $\|f\|_{L^2} \leq C_1 \|c^{-1/2}\alpha\|_{L^2}$

2) If (M,ω) is complete non-compact, the result still holds: $\exists\ \chi_R$ with compact support s.t. $\chi_R = 1$ on $B(p,R)$, $|\nabla\chi_R| \leq 1$

Given $\eta \in \Omega^{0,1}(L)$ with compact support,

$\eta = \eta_1 + \eta_2$, $\eta_1 \in \ker\bar\partial$, $\eta_2 \perp \ker\bar\partial$

$$|\langle \alpha, \eta \rangle| = |\langle \alpha, \eta_1 \rangle| \leq \| C^{-1/2} \alpha \|_{L^2} \left(\int_M (Ric_{j\bar{k}} + \Omega_{j\bar{k}}) \eta_{1,j} \overline{\eta_{1,\bar{k}}} \right)^{1/2}$$

$\forall R > 0$ $\displaystyle\int_M (Ric_{j\bar{k}} + \Omega_{j\bar{k}}) (\chi_R \eta_{1,j}) (\overline{\chi_R \eta_{1,\bar{k}}})$

$$\leq \| \bar{\partial} (\chi_R \eta_1) \|_{L^2}^2 + \| \bar{\partial}^* (\chi_R \eta_1) \|_{L^2}^2$$

$\bar{\partial}(\chi_R \eta_1) = (\bar{\partial} \chi_R) \eta_1 \longrightarrow 0$ in L^2

$\bar{\partial}^*(\chi_R \eta_1) = -(\nabla_j \chi_R) \eta_{1,\bar{j}} + \chi_R \bar{\partial}^* \eta_1 \to \bar{\partial}^* \eta_1$ in L^2 as $R \to \infty$

$$\Longrightarrow \quad |\langle \alpha, \eta \rangle| \leq \| C^{-1/2} \alpha \|_{L^2} \| \bar{\partial}^* \eta \|_{L^2} .$$

3) For version with $(n,1)$-forms, it is enough if M admits a complete Kähler metric $\tilde{\omega}$ even if (M, ω) is <u>not</u> complete.

<u>Idea</u>: Let $\omega_\varepsilon = \omega + \varepsilon \tilde{\omega}$

If $\Omega_L \geq c\omega$, then $\Omega_L \geq c_\varepsilon \omega_\varepsilon$, $c_\varepsilon > 0$

Given $\alpha \in \Omega^{n,1}(L)$, $\bar{\partial}\alpha = 0$

$\exists$ $f_\varepsilon \in \Omega^{n,0}(L)$ s.t. $\bar{\partial} f_\varepsilon = \alpha$,

$$\| f_\varepsilon \|_{L^2_\varepsilon} \leq \| c_\varepsilon^{-1/2} \alpha \|_{L^2_\varepsilon} \leq \| c^{-1/2} \alpha \|_{L^2}$$

needs calculation

can extract limit f as $\varepsilon \to 0$

E.g. If (M,ω) is Kähler (incomplete)

and $\exists\, \psi : M \to [0,1)$ s.t. $\sqrt{-1}\partial\bar\partial\psi \geq 0$

and $\{\psi \leq c\}$ are compact for $c<1$

then M admits a complete Kähler metric

(can take $\omega + i\partial\bar\partial\left(\frac{1}{1-\psi}\right)^2$)

Return to Donaldson-Sun's Thm

Argue by contradiction.

Suppose $(M_k^n, g_k, p_k) \xrightarrow[\text{pointed GH}]{} (C(X), 0)$

s.t. $\mathrm{Vol}(B(p,1)) > v$, s.t. $|\mathrm{Ric}(g_k)| < \frac{1}{k}$, $L_k \to M_k$ with curv. $\sqrt{-1}\Omega_{L_k} = \omega_k$

We will construct

a suitable section of L_k^m for suff. large k.

Special case: $C(X) = \mathbb{C}^n$, let $R>0$

$\exists\, \Phi_k : B(0,R) \hookrightarrow M_k$ for $k \gg 1$

$0 \longmapsto p_k$

s.t. $\Phi_k^* g_k \to g_{Euc}$ in $C^{1,\alpha}$

use Φ_k to identify $B(0,R)$ with its image.

J_k = complex structures of M_k

can assume $J_k(0) = J_{Euc}(0)$

J_k parallel $\Rightarrow$ $J_k \to J_{Euc}$ in $C^{1,\alpha}$

[reason: $\nabla(J_k - J_{Euc}) = \underbrace{(\nabla - \nabla_k)}_{\sim \nabla g_k} J_k$

$\Rightarrow \|\nabla(J_k - J_0)\|_{C^\alpha_{g_{Euc}}} \leq \Psi(k^{-1}) \|J_k\|_{C^\alpha_{g_{Euc}}} \leq \Psi(k^{-1})$]

<u>Claim</u> For $k \gg 1$ we have holo. charts $\{z^{(k)}_1, \dots, z^{(k)}_n\}$ on $B(0, R/2)$ w.r.t. J_k cx-structure, s.t.

$$\|z^{(k)} - z\|_{C^{2,\alpha}(B(0,R/2))} \leq \Psi(k^{-1})$$

<u>Pf of claim</u> Let $\varphi = \sum |z_i|^2$ on $B(0,R)$

since $J_k \to J_{Euc}$ in $C^{1,\alpha}$

$$\sqrt{-1}\, \partial_{J_k} \bar{\partial}_{J_k} \varphi > \frac{1}{2} \omega_k \quad \text{for } k \gg 1$$

$\Rightarrow$ can apply Hörmander estimate to L = trivial bundle over $(B(0,R), J_k, \omega_k)$ with metric $e^{-\varphi}$

$$Ric(\omega_k) + \sqrt{-1}\Omega > \frac{1}{4}\omega_k$$

~~we have trivia~~